Brainstorming and Ideation

Date	
Team ID	
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

A team brainstorming session was conducted to identify solutions for challenges in blood pressure monitoring and early risk detection. Various ideas were generated, including a BP tracking system, machine learning-based prediction model, health alerts, and data visualization dashboard.

The ideas were grouped and evaluated based on feasibility, impact, and innovation. After analysis, the team selected a Machine Learning-based Blood Pressure Risk Prediction System as the core solution due to its potential for early detection and proactive healthcare management. The generated ideas were evaluated based on impact, feasibility, and technical complexity. The Machine Learning-based Blood Pressure Risk Prediction System was prioritized as it offers high impact and practical implementation.

1 Problem Understanding

Before developing Predictive Pulse, a detailed understanding of the problem was conducted:

- Hypertension is a “silent killer” often going undiagnosed until complications arise.
- Early detection is crucial for preventive care.
- Traditional methods require frequent doctor visits and manual record-keeping.
- A digital solution using machine learning can automate risk prediction and provide clinical recommendations.

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2 Ideation Process

The ideation process involved generating ideas to design an intelligent blood pressure analysis system:

1. **Feature Identification**
    - Identify critical features that influence hypertension: Age, Gender, Systolic BP, Diastolic BP, Family History, Treatment Status, Lifestyle Factors.
2. **Data Source Exploration**
    - Brainstormed sources of reliable data, such as public medical datasets, hospital records, and synthetic datasets for model training.
3. **Algorithm Selection**
    - Explored different ML algorithms:
        - Logistic Regression → Simple and interpretable
        - Decision Tree → Handles categorical data well
        - Random Forest → Reduces overfitting, higher accuracy
4. **System Architecture Ideas**
    - Considered a web-based application for accessibility.
    - Flask framework chosen for easy integration with Python ML models.
    - HTML/CSS/Bootstrap for professional UI.
5. **Risk Classification Concept**
    - Brainstormed a system to classify hypertension into stages: Normal, Elevated, Stage 1, Stage 2.
    - Included confidence score and color-coded risk levels for clarity.
6. **Validation and Security**
    - Real-time form validation for correct inputs.
    - Safe data handling to protect patient information.

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3 Innovative Features Considered

- **Color-Coded Risk Levels** – To quickly identify high-risk patients.
- **Clinical Recommendations** – Advice based on predicted hypertension stage.
- **Responsive Design** – Accessible on mobile and desktop.
- **Data Storage for Future Analysis** – Option to save patient history.
- **Scalable Architecture** – Ready for future AI integration (e.g., wearable device data).

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4 Selection of Final Idea

After brainstorming multiple possibilities, the team decided to:

- Build a **Random Forest-based model** for better accuracy.
- Deploy it using **Flask Web Application** for real-time risk prediction.
- Focus on **simplicity and medical-grade design** for usability.
- Include **visual feedback** (confidence score, color-coded stages) to enhance user experience.